Reduction and calibration of spectra

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This project was focused on processing raw data collected by Professor Alberto Rebassa using the ISIS instrument on the William Herschel Telescope in La Palma. We employed specialized astronomical software (Starlink, molly, pamela) to conduct bias reduction, flat field correction, and sky correction. Additionally, we successfully generated an arc-calibrated spectrum for both blue and red parts of the spectra.

I. INTRODUCTION

Spectroscopic observations provide crucial insights into celestial objects, yet raw data often contain noise and artifacts, impeding accurate analysis. This project aims to refine data reduction techniques for spectral data acquired from astronomical observations. By employing advanced computational software (starlink, pamela and molly), we seek to improve data quality by mitigating noise, correcting instrumental effects, and optimizing extraction methods.

The data used in this project was taken by Professor Alberto Rebassa in ING la Palma in Canarias where is located the William Hershel telescope. This is a reflector telescope with 4.2 meters in diameter designed to detect infrared and visible light [2]. The instrument used is ISIS. The Intermediate-Dispersion Spectrograph and Imaging System (ISIS) offers high-resolution capabilities for both imaging and spectroscopy across various wavelength ranges, enabling detailed studies of celestial objects[1]. The detector employed is EEV12 a CCD (charge couple device) that has a high quantum efficiency but some issues such as uniformity, the sensitivity varies from pixel to pixel, dark current, bias level which is the generated mark for no incoming photons with exposure time zero, and flat field image. The dark current effects are minimized by cooling the CCDs with liquid nitrogen. However, for the intrinsic zero level of the CCD we use either bias images and overscan regions to calibrate raw images[3].

Specifically, our goal for the data reduction is to carry out a bias reduction, flat field correction, and sky correction. Once we achieve clean spectra we obtain an arc-calibrated spectrum, crucial for precise determination of spectral lines and accurate wavelength assignment.

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II. DATA REDUCTION

A. Blue

1. *Bias subtraction*

The initial phase of the project revolves around data reduction processes. The bias image serves to establish the baseline or zero level across the entire CCD camera, effectively eliminating electronic noise inherent in the detector. Additionally, overscan data is gathered for each spectrum captured throughout the night, providing a means to monitor and detect any bias level fluctuations. The procedure involves combining and subtracting bias images to effectively calibrate and remove electronic noise from the acquired spectra.

For the blue spectra, we follow the values given by the professor. Consequently, we choose to detect the over-scan region between 60 and 300 on the x-axis and between 4120 to 4200 on the y-axis.

2. *Flat field correction*

Flat field correction in astronomy is a crucial image-processing step aimed at removing non-uniformities or variations in sensitivity across an image. It involves the division of an acquired image by a flat field image to correct for uneven illumination, dust on the optics, detector imperfections, or any other systematic variations in the sensitivity of the imaging system.

After subtracting the bias from the images and ensuring that the function's tail reaches zero, we followed the guidelines specified in the flatfield file. Initially, we merged all flat images. However, as we dealt with a two-dimensional image, we condensed the x-axis into a single line. To prevent abrupt transitions at the extremes, the Professor delimited an x-axis range between 10 and 480, also the y-axis was delimited between 140 to 3750. Subsequently, we applied a logarithmic transformation to facilitate function fitting, adhering to the professor's recommendation of employing a 7th-order polynomial to best adjust the function.

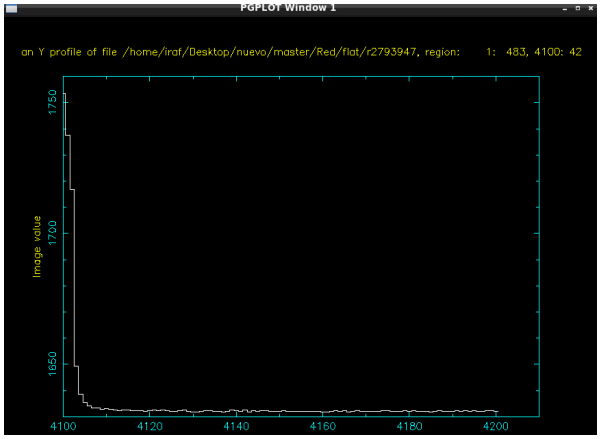


Figure 1. Pplot between the established limits (x: 10-480, y: 140-3750) of the spectrum and the 7th-order polynomial fit. As we see it matches good

3. *Sky correction*

Sky correction in astronomy involves removing or mitigating the effects of the sky's background signal from astronomical observations.

In this case we did not edit neither the noise file nor the data file. The values in the noise file were the same as the showed in class. For the reference spectrum image (r2794030.sdf). Moreover, the horizontal position of the spectrum (xpos= 260), his range of possible fluctuation (230-290). And, again the y-axis region chosed previously in the flat field correction(480-3750).

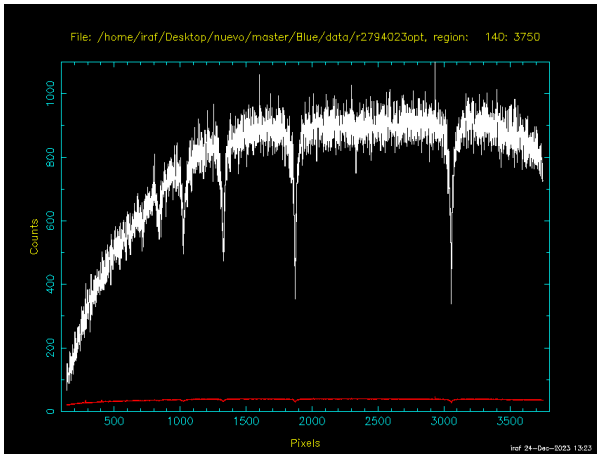


Figure 2. Pplot of r2794023opt.sdf, one of the five final optimal files returned.

B. Red

1. *Bias subtraction*

First, we look for the over-scan region. To find this region we plot any profile from flatfield folder. Making zoom in on the tail of this picture we realize that values fall dramatically to a constant number(bias level), this is our over-scan region. In other words, The limits of the overscan region are established when the background value (bias level) stabilizes, as pictured in figure 3.

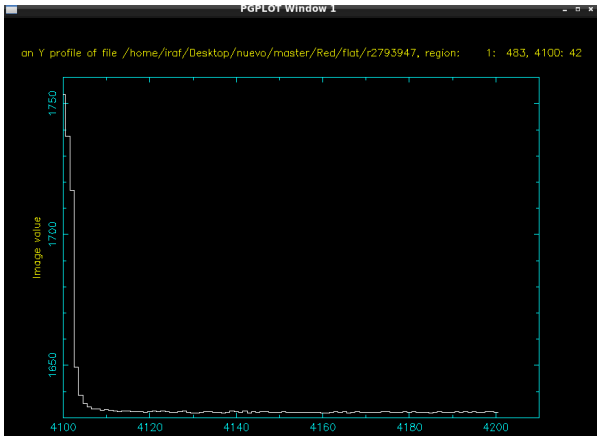


Figure 3. A pplot of r2793947.sdf making zoom on the tail of the image. We can subtract the over-scan limits from a quick analysis of the picture. Boundaries are 4120 to 4200 in y-axis

Additionally, we observed that the x-axis required adjustments as well. In a plot along this axis, it became apparent that the lower pixels exhibited values lower than the average. Hence, we excluded these values, but also higher ones to agree with the flat analysis explained in section II B 2. Boundaries taken are 30 to 320 on x-axis.

Finally, we used the command makebias in which we input a cluster number of three. It means that the first average between the biased images was made by combining three images.

2. *Flat field correction*

The main important part of this section is to modify the flatfield file. This file contains the procedure described previously in section II A 2. First, we had to modify the section in the x range which will be collapsed. This instruction is in the second line of the flatfield file.

In image 4 we notice that, as in the blue part, the beginning and the final present no-soft functions. Moreover, this red data also presents an aberration in the

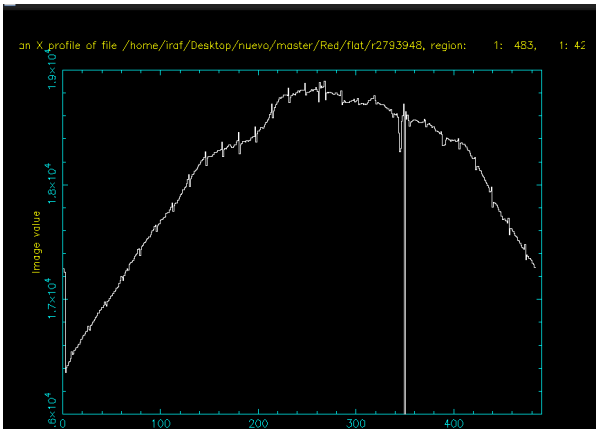


Figure 4. Full pplot of the flat file along the x-axis, that represents pixels as usual. From this image, we determined the x range to use in the collapse of the spectrum. The plot exhibits irregularities in the first pixels and around 350 (abrupt fall)

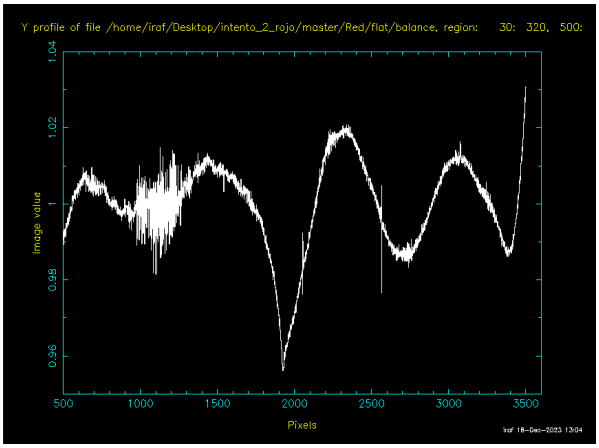


Figure 6. Pplot of the balance.sdf file in the selected region 500 to 3500 and 30 to 320. This depiction serves to evaluate the whole process. The function around one with tiny variations

middle of the plot, a single pixel(around 350) that drops abruptly. In conclusion, we cutted more than in the blue data, defining the x collapsing range between 30 and 320.

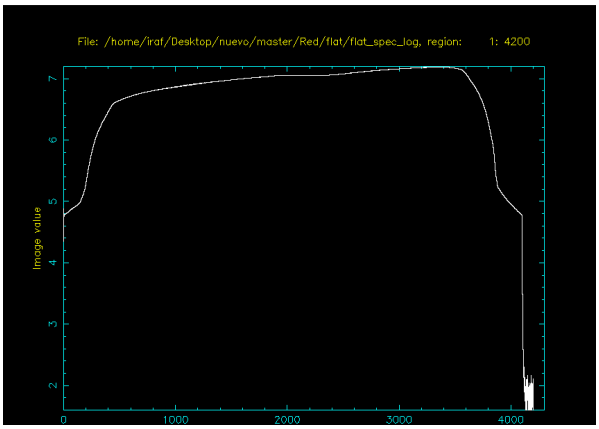


Figure 5. Full pplot of flatspcelog file. The function is soft in almost the whole range. However, it exhibits irregularities and sharp changes in the slope at the beginning and in the final. From this picture, we determined the y range in which the fit. is made

To avoid sharp angles shown in 5 and following the procedure made in the blue part we chose the boundaries in 500 and 3500. These limits can be a little bit restrictive, but we presume to have enough data.

Due to the extra curvature in this function, a higher-order polynomial fit was selected. Instead of seven, eight.

To verify the confidence in our results, the depiction 6 serves as a significant indicator. The nearly constant function around one, with minimal fluctuations, suggests a strong correlation between the fit and the spectrum.

C. Sky correction

As in the flat field correction subtracting the bias from each one of the spectrum images is the first step of the sky correction.

Next, the values in the noise file were changed. Specifically, we edited the reference spectrum image(r2794016.sdf). Consequently, the horizontal position of the spectrum (xpos= 247), his range of possible fluctuation (217-277). And, again the y-axis region in which we work all along this project (500-3500). Once these changes are made noise file is executed.

Other important values returned after the sky reduction were the mean sky, 133.464, the object position, 253, and the Chi-squared of fit, 1.112. The last one is an indicator of the error in the sky correction, due to this value being close to 1, the profile fit is good enough.

Finally, we execute "reduce datafile" to get the final spectrum images optimized.

To evaluate the effectiveness of the reduction process, we employ the command "loganal zzzreduction.log", which provides a range of error indicators. The skyfit outcomes reveal an outstanding average RMS value of 1.0, signifying excellent performance. Additionally, the profit results display chi-squared values of 1.13, 1.15, 1.24, 1.77, and 1.52.

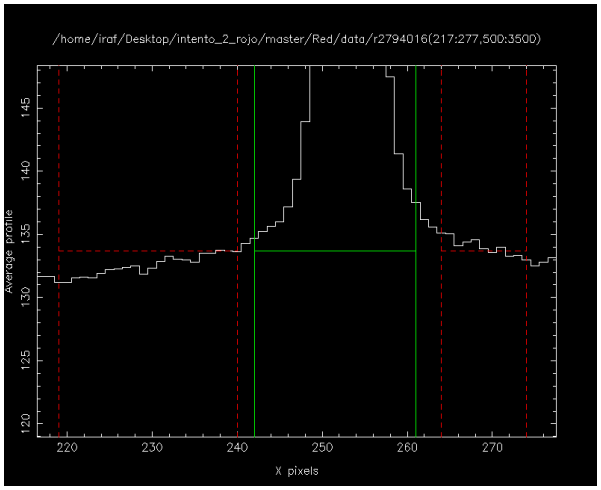


Figure 7. Picture plotted by the noise file. Red vertical lines enclose sky regions, while vertical green lines enclose star region. The average RMS of this enclosed was 1.01

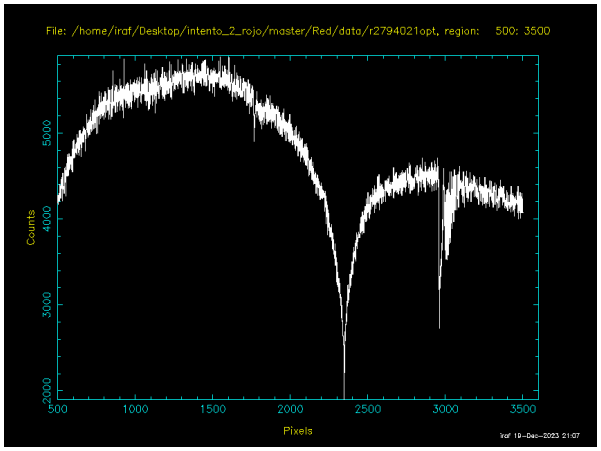


Figure 8. Pplot of r2794021opt.sdf, one of the five final optimal files returned after the three steps bias reduction, flat field correction ,and sky correction

III. WAVELENGTH CALIBRATION

Upon obtaining the optimal reduced spectrum, the subsequent step involves delving into the interpretation of the image's physical implications. However, before this analysis, a crucial initial step is to transform the axis quantities into physical quantities. It means to convert pixels and counts to wavelength and flux. This transformation requires establishing the relationship between pixel values and wavelengths by comparing theoretical spectra with our obtained data. In order to do this we relied in the astronomic software Molly

A. Blue

As a result of setting the initial threshold at 60000 counts the software successfully discerned 30 spectral lines, surprisingly, two extra lines than in the instruction video. We accepted all the lines even though the numbers of the lines between the video[4] and our analysis vary because of the extra two lines.

Furthermore, to augment our dataset, well-known spectra of argon, helium, and neon were incorporated from the provided data folder. This supplemental addition of established spectra enriched our analysis and expanded the reference pool for comparison and validation purposes.

To enhance our examination of the spectral data, the CuNe+CUAR spectrum from the Blue pdf was meticulously scrutinized. In this part, we followed the analysis made by Professor Alberto Rebassa [4], paying special attention to the selected lines because of the difference in the number of lines.

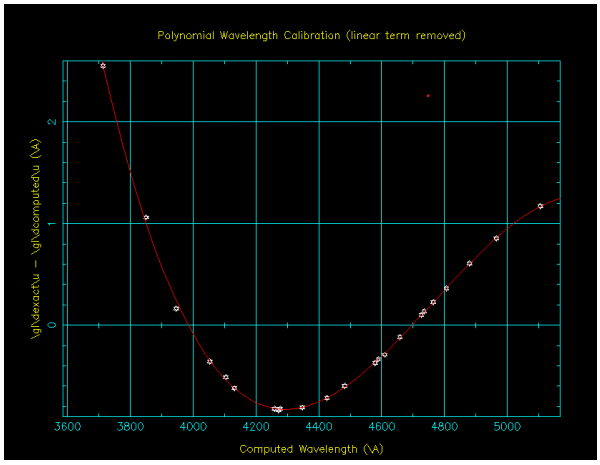


Figure 9. Profit achieved at the final of the line-identification process in the blue part. This image shows a really precise fit that matches most of the values, even those in the extremes of the graph. This fit will be used to calibrate the other spectra

Now we have an arc-calibrated for one of the spectra, it is easy to apply the same fit to the five arcs making use of molly with the same order of fit (5) but with 2 low order variable coefficients(tweak). These arcs calibrated are used to calibrate the spectra with a method "N", which means the nearest arc in time for each spectrum. The final calibrated spectra are as follows in chart10

B. Red

We are now moving on to processing the red spectra with the same reduction steps. During this phase, the

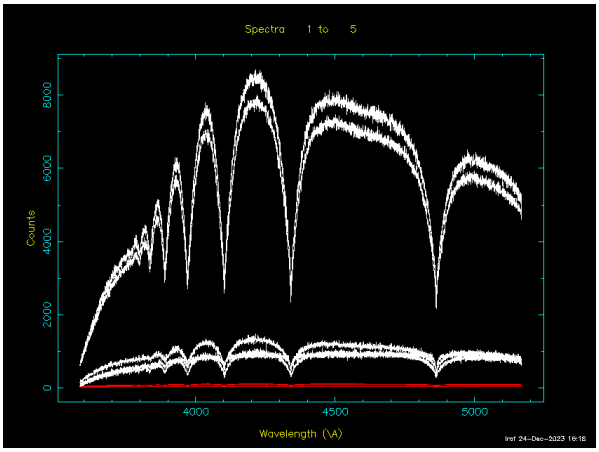


Figure 10. Plot of the blue spectra (5 spectrum images) after the reduction process and wavelength calibration in which we can identify the absorption lines

program pinpointed 28 lines by employing a threshold of 6000. In the red.pdf we search for the CuNe+CuAr table centered in 6749 angstroms because our arcline spectrum is centered in 6400.0 angstroms.

The first line that was identified was the pixel 1519 with the wavelength 6402.25. We chose this line because the local patron is very similar and it is located in the center of the spectrum. This made easier its detection.

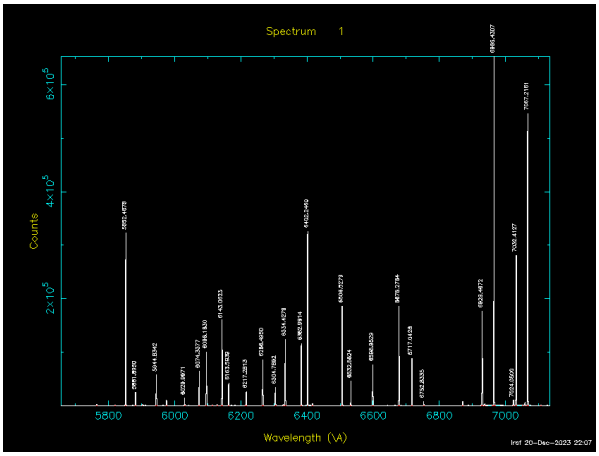


Figure 11. Plot of the lines identified by the software, as we can see, most of the lines have already an associated wavelength used to improve the fit

As in the blue part, in the beginning, we identified the lines in the center of the spectrum. Some of those central lines are pixel 1480 wavelength 1482 and pixel 1731 wavelength 6506. The fit was improving line by line, at first with a single line then a square fit, and with about nine lines finally, we change the polynomial order to 5 and got a proper fit.

In this way, we reach a good enoguh profit to predict the value of the wavelegnth given a pixel. However, the values predicted were more precise in the center, as a result we start form this part to the borders of the function. As result we obtained depiction 12.

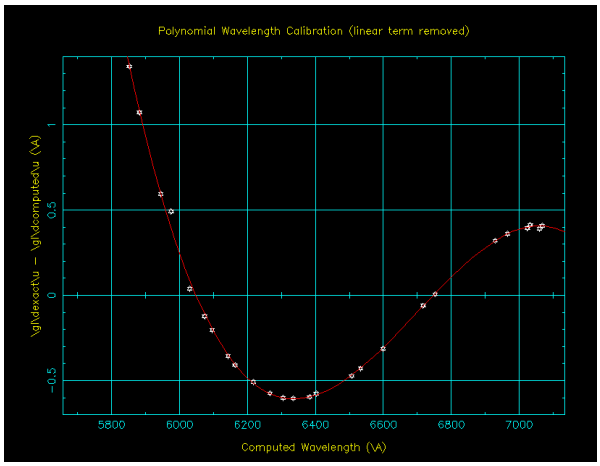


Figure 12. Profit achieved at the final of the line-identification process. This image shows a really precise fit that matches most of the values, even those in the extremes of the graph. This fit will be used to calibrate the other spectra

Once we finished the arc-calibration, we applied to the rest of the arcs and we used the arcs to calibrate the spectra with the following the command in "wave cal" file. Finally, we got the spectra with the wavelength in the x-axis 13.

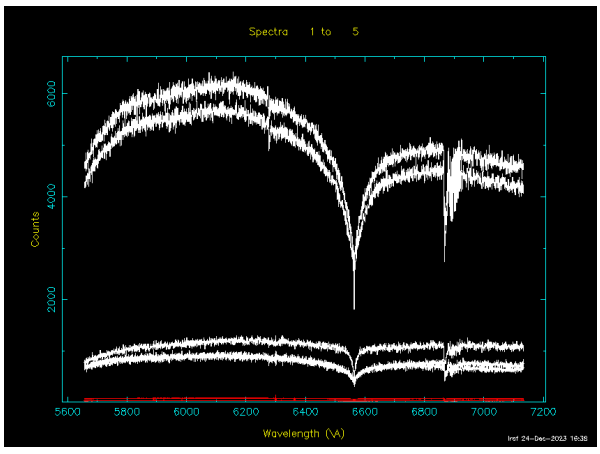


Figure 13. Plot of the red spectra (5 spectrum images) after the reduction process and wavelength calibration in which we can identify the absorption lines

IV. RESULTS

Finally, we should remember that telescope William Hershel and specifically the instrument ISIS has two filters to divide the spectrum into blue and red parts. After merging the two parts we present the reduced arc-calibrated spectra in which the left part of the spectra (short wavelengths) shows the process carried out in the blue sections, and in the right of the spectra(large wavelengths) the results of the process carried out in the red sections. The merged spectrum are plotted in images 14,15,16,17,18.

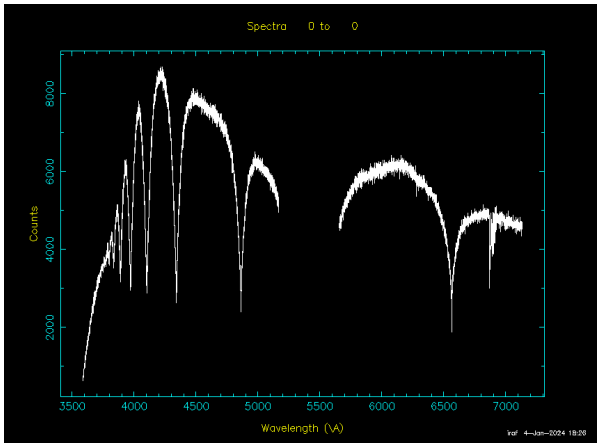


Figure 14. Plot of the full arc-calibrated and reduced spectrum of the object SP0943+441. The x-axis is in Angstroms and the y-axis is in counts.

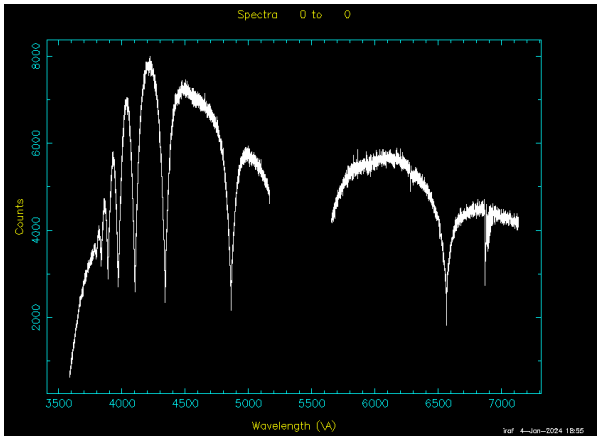


Figure 15. Second plot of the full arc-calibrated and reduced spectrum of the object SP0943+441. The x-axis is in Angstroms and the y-axis is in counts.

V. CONCLUSIONS

In this study, our primary focus was to process and refine raw astronomical data obtained by Professor Alberto Rebassa using the ISIS instrument on the William Herschel Telescope in La Palma. Leveraging specialized astronomical software such as Starlink, Molly, and Pamela, our objectives centered on implementing robust data reduction techniques.

Through a meticulous approach, we successfully executed bias reduction, flat field correction, and sky correction procedures. These critical steps aimed to eliminate noise, rectify instrumental effects, and optimize extraction methods. The culmination of these efforts yielded an arc-calibrated spectrum, a pivotal outcome facilitating precise spectral line determination and accurate wavelength assignment.

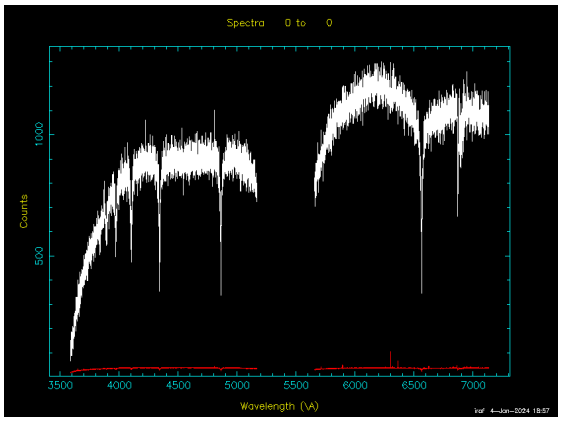


Figure 16. Plot of the full arc-calibrated and reduced spectrum of the object J0820+3834. The x-axis is in Angstroms and the y-axis is in counts.

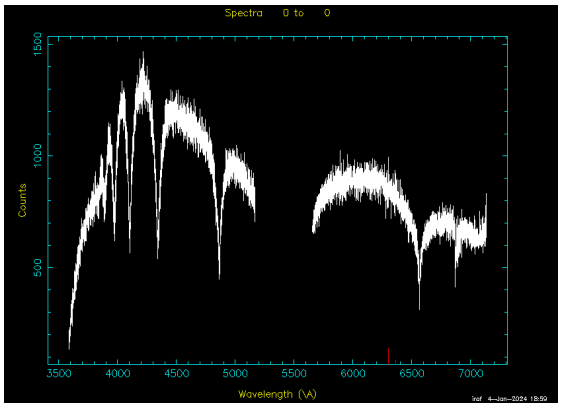


Figure 17. Plot of the full arc-calibrated and reduced spectrum of the object J0901+1657. The x-axis is in Angstroms and the y-axis is in counts.

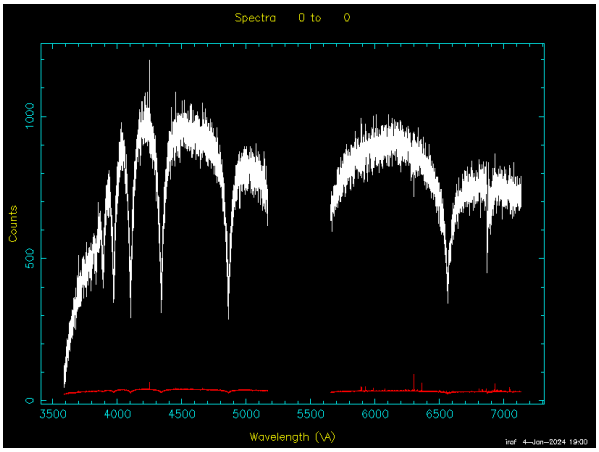


Figure 18. Plot of the full arc-calibrated and reduced spectrum of the object J0907+2733. The x-axis is in Angstroms and the y-axis is in counts.

[1] Ing (2021). ISIS. [Online; accessed 23. Dec. 2023].
[2] Ing (2023). William Herschel Telescope. [Online; accessed 23. Dec. 2023].
[3] Rebassa, A. (2022). Telescopes and observational techniques. [Online; accessed 23. Dec. 2023].
[4] Rebassa, A. (2023). Computational Astrophysics (2021-03-23 at 06:05 GMT-7). [Online; accessed 24. Dec. 2023].